SQL Project - Netflix Data Analysis

Note: PFA database file, 'netflix database.sql', run it in your workbench to create the database.

Analysis Report:

1. Using the Viewing History table, identify the top 3 most-watched movies based on viewing hours.

```
select content.TitleName , count(viewinghistory.ContentID) as total_Views  ,
sum(viewinghistory.Runtime) as Total_Viewing_Hours from
content join viewinghistory
on content.ContentID =viewinghistory.ContentID
group by content.TitleName
order by Total_Viewing_Hours desc limit 3;
```

Result Output -

TitleName	total_Views	Total_Viewing_Hours
Spider-Man	4	128
Titanic	3	107
Extraction	3	64

2. Partition the viewing hours by category and genre to find the top genre in each category. Use the rank function to rank genres within each category.

```
 with ranked_genres as (
 select
   c.category,  c.genre,
   sum(v.runtime) as total_viewing_hours,
   rank() over (partition by c.category order by sum(v.runtime) desc) as genre_rank
from content c
join viewinghistory v
on c.contentid = v.contentid
group by c.category, c.genre
)
select category, genre, total_viewing_hours
from ranked_genres
where genre_rank = 1
order by category;
```

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Result Output –

Category	Genre	Total_Viewing_Hours
Movie	Action	205
TV Show	Crime	60

3. Determine the number of subscriptions for each plan. Display Plan ID, Plan Name and Subscriber count in descending order of Subscriber count.

```
Select
plans.PlanID ,plans.PlanName, Count(subscribes.CustID) as Number_of_subscription
from plans
join subscribes on plans.PlanID = subscribes.PlanID
group by plans.PlanID, plans.PlanName
order by Number_of_subscription desc;
```

Result Output -

PlanID	PlanName	Number_of_subscription
P2	Standard	16
P1	Standard w/ Ads	14
P3	Premium	13

4. Which device type is most commonly used to access Netflix content? Provide the Device Type and count of accesses.

```
select devices.DeviceType,count(uses.DeviceID) as Total_No_Device  from uses
join devices
on uses.DeviceID = devices.DeviceID
group by devices.DeviceType
order by Total_No_Device desc;
```

Result Output -

DeviceType	Total_No_Device
Tablet	14

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5. Compare the viewing trends of movies versus TV shows. What is the average viewing time for movies and TV shows separately?

```
select  content.Category ,
COUNT(viewinghistory.ContentID) AS Total_Views,
avg(viewinghistory.Runtime) as Avg_Total_Viewing_Hours from
content
join viewinghistory
on content.ContentID = viewinghistory.ContentID
group by content.Category
order  by Avg_Total_Viewing_Hours desc ;
```

Result Output –

Category	Total_Views	Avg_Total_Viewing_Hours
TV Show	3	35.6667
Movie	14	28.0714

6. Identify the most preferred language by customers. Provide the number of customers, and language.

```
select customerslanguagepreferred.Language , count(customers.CUSTID ) as
Total_Number_OF_Customers from
customerslanguagepreferred join
customers on customerslanguagepreferred.CustID = customers.CUSTID
group by customerslanguagepreferred.Language
order by  Total_Number_OF_Customers desc  limit 1;
```

Result Output -

Language	Total_Number_OF_Customers
English	46

7. How many customers have adult accounts versus child accounts? Provide the count for each type.

```
select   'Adult' as Type_Of_Customer,COUNT(adultacc.ProfileID) AS
Total_Accounts
from adultacc
          union all
select 'Child' as Type_Of_Customer , count(childacc.ProfileID) as Total_Accounts
from childacc ;
```

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Result Output –

Type_Of_Customer	Total_Accounts
Adult	10
Child	7

8. Determine the average number of profiles created per customer account.

```
select
avg(profile_count) AS avg_profiles_per_customer
from (
select
count(*) AS profile_count
from profiles
group by CustID
) as t;
```

Result Output –

avg_profiles_per_customer
2.8333

9. Identify the content that has the lowest average viewing time per user. Provide the titles and their average viewing time.

```
select content.TitleName,
  avg(viewinghistory.Runtime) as Avg_Viewing_Time
from content
join viewinghistory
on content.ContentID = viewinghistory.ContentID
group by content.TitleName
order by Avg_Viewing_Time asc limit 1 ;
```

Result Output -

TitleName	Avg_Viewing_Time
Hulk	13

10. Determine the count for each content type.

```
select Category as Each_Content_Type , count(ContentID) from content
group by Category;
```

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Result Output –

Each_Content_Type	count(ContentID)
Movie	23
TV Show	7

11. Compare the number of customers that have unlimited access and who do not.

```
Select case
when plans.ContentAccess = 'unlimited' then 'Unlimited Access'
else 'Limited/Standard Access'
end as AccessType,
subscribes.Status,
count(distinct subscribes.CustID) as Number_of_Customers
from
Plans
join    Subscribes  ON plans.PlanID = subscribes.PlanID
group by AccessType, subscribes.Status;
```

Result Output -

AccessType	Status	Number_of_Customers
Limited/Standard Access	Active	11
Limited/Standard Access	Inactive	3
Unlimited Access	Active	17
Unlimited Access	Inactive	12

12. Find Average monthly price for plans with Content Access as "unlimited".

```
SELECT  avg(plans.MonthlyPrice) as Avg_Monthly_Price_Unlimited_Access
from plans
where ContentAccess = 'Unlimited';
```

Result Output –

Avg_Monthly_Price_Unlimited_Access
19.24

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13. List all the customers who have taken the plan for till 2028 and later. Display CustomerID, Customer name and Expiration Date of the plan, ordered by Expiration Date in descending order first, and then by Customer Name.

    Notes:- I use 2025 in the WHERE clause because, in the current dataset, no plans exist after 2025.

```
select
C.CUSTID AS CustomerID, concat( C.FNAME,' ',C.LNAME) as  customer_name,
S.EndDate
from
Customers C
join
Subscribes S on C.CUSTID = S.CustID
where
year(S.EndDate) >= 2025
order by S.EndDate desc , customer_name asc;
```

Result Output -

CustomerID	customer_name	EndDate
C15	James Martin	28/02/2025
C51	Samantha Gomez	28/02/2025

14. Display Average Revenue generated from each city. Rank city based on average revenue.

```
select
paymentmethod.City,
avg(paymenthistory.PaymentAmount)  as  Average_Revenue,
dense_rank() over (order by avg(paymenthistory.PaymentAmount) desc) as
Revenue_Rank
from  paymentmethod
join paymenthistory on  paymentmethod.CardID = paymenthistory.CardID
group by  paymentmethod.City
order by Revenue_Rank;
```

Result Output –

City	Average_Revenue	Revenue_Rank
Salt Lake City	22.99	1
Boulder	22.99	1
Cody	15.49	2
Austin	15.49	2
Denver	6.99	3

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15. Display most frequently viewed genre among adults for each category.

```
with Ranked as (SELECT
content.Category as Category,
content.Genre as Genre,
sum(viewinghistory.Runtime) AS most_Freq_view_genreby_Adult_,
rank() over(partition by content.Category order by sum(viewinghistory.Runtime) desc) as
 Most_Rank
from
adultacc
  join viewinghistory on adultacc.ProfileID = viewinghistory.ProfileID
  join content on  content.ContentID = viewinghistory.ContentID
group by Category , Genre
)
select * from Ranked
where Most_Rank = 1
order by Category ;
```

Result Output -

Category	Genre	most_Freq_view_genreby_Adult_	Most_Rank
Movie	Action	96	1
TV Show	Crime	60	1

********** **The End** **********

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